

15.7 Two photons are initially prepared at a source and move along the y -direction in opposite directions toward polarisers at A and B. They are initially in the state

$$|\Psi\rangle = \frac{1}{\sqrt{2}}[|V_A V_B\rangle + |H_A H_B\rangle],$$

where $|V_I\rangle$ and $|H_I\rangle$ are the two linear polarisation states, vertical and horizontal, of the photon moving to the polariser at $I = A$ or B . To be explicit, we will fix the vertical polarisation direction here as the $\hat{z}$ -direction. In particular, the V state is in the direction of $\hat{z}$ and the H state is in the $\hat{x}$ -direction, thus perpendicular to both the $\hat{z}$ -direction and the direction of motion for the photon. Let $\underline{n}_I$ be other directions at polarisers $I = A$ or B that are perpendicular to the direction of motion of the photons, so in the $\hat{x}$ - $\hat{z}$ plane, and at an angle θ_{Az} to the $\hat{z}$ -direction. If photon I is measured through a polariser oriented with respect to the $\underline{n}_I$ -direction:

$$|V_{I,I}\rangle = \cos \theta_{Iz} |V\rangle - \sin \theta_{Iz} |H\rangle,$$

$$|H_{I,I}\rangle = \sin \theta_{Iz} |V\rangle + \cos \theta_{Iz} |H\rangle.$$

The two photons initially in the state $|\Psi\rangle$ are then measured respectively at A and B, with polarisers oriented along the $\underline{n}_A$ and $\underline{n}_B$ -directions, respectively. Let $P_{VV}(\underline{n}_A, \underline{n}_B)$ be the probability that the vertical polarisation of the photon at A is measured along direction $\underline{n}_A$ for polarisers rotated by angle θ_{Az} and for the photon at B the vertical polarisation is measured along direction $\underline{n}_B$ for polarisers rotated by angle θ_{Bz} . Show that

$$P_{VV}(\underline{n}_A, \underline{n}_B) = |\langle V_{A,A} V_{B,B} | \Psi \rangle|^2 = \frac{1}{2} \cos^2(\theta_{B1} - \theta_{A1}).$$

Also compute $P_{HH}(\underline{n}_A, \underline{n}_B)$, $P_{VH}(\underline{n}_A, \underline{n}_B)$ and $P_{HV}(\underline{n}_A, \underline{n}_B)$.

Show that

$$\begin{aligned} E(\underline{n}_A, \underline{n}_B) &\equiv P_{VV}(\underline{n}_A, \underline{n}_B) + P_{HH}(\underline{n}_A, \underline{n}_B) - P_{VH}(\underline{n}_A, \underline{n}_B) - P_{HV}(\underline{n}_A, \underline{n}_B) \\ &= \cos[2(\theta_{Az} - \theta_{Bz})]. \end{aligned}$$

For the expression

$$S = E(\underline{n}_A, \underline{n}_B) - E(\underline{n}_A, \underline{n}'_B) + E(\underline{n}'_A, \underline{n}_B) + E(\underline{n}'_A, \underline{n}'_B),$$

a realistic local theory gives the Bell inequality $-2 \leq S \leq 2$. Show that this inequality is violated for the quantum-mechanical expressions for photons derived here. For this choose the orientations

$$\theta_{Bz} - \theta_{Az} = \theta_{A'z} - \theta_{Bz} = \theta_{B'z} - \theta_{A'z} = \phi \quad \text{and} \quad \theta_{B'z} - \theta_{Az} = 3\phi,$$

$$\underline{n}_i \cdot \underline{n}_j = \underline{n}'_i \cdot \underline{n}_j = \underline{n}'_i \cdot \underline{n}'_j = \cos \phi \quad \text{and} \quad \underline{n}_i \cdot \underline{n}'_j = \cos 3\phi$$

and plot as a function of ϕ . Determine the range of angles for ϕ for which Bell's inequality is violated.